

portions of the audio can be discarded. MP3 audio can be compressed with different bitrates, providing a range of tradeoffs between data size and sound quality. Typically, rates chosen are between 64 and 320 kbps. (Remember that uncompressed audio as stored on a CD has a bitrate of 1411.2 kbps (16 bits/sample x 44100 samples/second x 2 channels). With too low a bit rate, “compression artefacts”—sounds that were not present in the original recording) may appear in the reproduction.

MP3 Surround, a version of the format supporting 5.1 channels for surround sound, was introduced in December 2004. MP3 Surround is backward compatible with standard stereo MP3, and file sizes are similar.

WMA

Windows Media Audio files are supposed to offer the same audio clarity level when you encode a piece of music at 96 kbps that MP3 does when it is encoded at 128 kbps. WMA is a proprietary compressed audio file format developed by Microsoft. It was initially a competitor to the MP3 format, but with the introduction of Apple’s iTunes Music Store, it has positioned itself as a competitor to the Advanced Audio Coding format used by Apple. WMA is part of the Windows Media framework. WMA is second only to MP3 in popularity in terms of number of devices supported.

AAC

Advanced Audio Coding (AAC) is also known as MPEG-2 Part 7. It was popularized by Apple through its iPod and iTunes Music Store. AAC was designed as an improved-performance codec relative to MP3.

Apple is doing a lot to promote AAC, and here’s a snippet from Apple’s site about the format:

- AAC provides audio encoding that compresses much more efficiently than older formats, such as MP3, yet delivers quality rivalling that of uncompressed CD audio.